



Rajshahi University of Engineering & Technology

Department of Mechatronics Engineering

MTE LAB REPORT – 01

Course No.: **MTE 3202**

Course Title: **Power Electronics and Drives Sessional**

Submitted By:

Abrar Faiyaz Anan Chowdhury
ID: **2008008**
3rd Year, Even Semester
Department of Mechatronics Engineering
Rajshahi University of Engineering &
Technology, Rajshahi-6204.

Submitted To:

Md. Firoj Ali
Assistant Professor
Department of Mechatronics Engineering
Rajshahi University of Engineering &
Technology, Rajshahi-6204.

Sarafat Hussain Abhi
Assistant Professor
Department of Mechatronics Engineering
Rajshahi University of Engineering &
Technology, Rajshahi-6204.

Date of Experiment: December 09, 2024

Date of Submission: January 06, 2025

Title

The Role of Power Electronics for Environmental Sustainability.

Abstract

Environmental sustainability has become a critical global concern, with the need to transition to cleaner energy sources and reduce carbon emissions being more urgent than ever. This report investigates the pivotal role of power electronics in achieving environmental sustainability. Power electronics, involving the efficient conversion, control, and management of electrical energy, is essential for integrating renewable energy sources, improving energy efficiency, and enabling sustainable transportation. This report will provide an overview of power electronics, its key features, and its applications in various sectors. It will also analyse relevant data, discuss future prospects, and present a perspective on the importance of power electronics in the context of Bangladesh's power sector. The findings highlight that power electronics is not just a technological advancement but a crucial driver for a sustainable future.

Introduction

Global warming, driven by the increasing concentration of greenhouse gases in the atmosphere, poses a significant threat to the planet. The International Panel on Climate Change (IPCC) has confirmed that human activities are the primary cause of the observed warming trend [1]. The consequences of inaction are severe, including rising sea levels, extreme weather events, and ecological disruptions. To mitigate these effects, a transition to sustainable energy sources and the adoption of energy-efficient technologies are imperative.

Sustainable energy, derived from renewable sources like solar, wind, hydro, and geothermal, offers a viable solution to reduce our dependence on fossil fuels [2]. However, the intermittent nature of these sources necessitates advanced technologies for efficient energy conversion and grid integration. Power electronics emerges as a critical technology in this regard. It is the study and application of solid-state electronics for the conversion and control of electric power. Its importance lies in its ability to enhance the efficiency of energy use, facilitate the integration of renewable energy sources, and support the development of sustainable transportation systems [3]. Power electronics can withstand high power, with less current consumption, and controllability. It has been estimated that roughly 15%-20% of electricity. According to estimates, the extensive use of power electronics can reduce electricity consumption by between 15 and 20 percent.

This report will delve into the role of power electronics in promoting environmental sustainability. It will explore the key features of power electronics, provide real-world examples and data, examine future trends, and discuss the specific implications for Bangladesh's power sector.

Key Features of Power Electronics

Power electronics systems offer several key features that contribute to environmental sustainability.

High Power Electronics: Advances in high-power electronics, particularly using wide-bandgap materials, are revolutionizing the electrical power system. These devices can operate at higher temperatures and voltages, leading to increased efficiency and reduced losses in power transmission and distribution.

Energy Efficiency: Power electronics enables significant energy savings across various applications. By precisely controlling the speed of motors, pumps, and fans, power electronics optimizes energy consumption in industrial processes and building systems. Hybrid and electric vehicles, powered by power electronics, offer substantial reductions in fuel consumption and emissions.

Integration of Renewable Energy: Power electronics plays a vital role in integrating intermittent renewable energy sources, such as solar and wind, into the electrical grid [4]. Power converters adjust the voltage and frequency of renewable energy output to match grid requirements, ensuring stable and reliable power supply.

Flexible Control: Power electronics provides advanced control capabilities for managing complex energy systems. This includes load balancing in smart grids, optimizing energy storage, and enabling the integration of distributed generation sources.

More Flexible Control by Power Electronics: Power electronics essentially employs high-efficiency switching power semiconductors to convert and control electrical power using dc-to-dc, dc-to-ac, ac-to-dc, and ac-to-ac converters, which are widely used in industrial, commercial, residential, transportation, aerospace, military, and utility systems.

Data and Applications

The positive impact of power electronics on environmental sustainability is evident in various sectors.

Smart Grids: Smart grids utilize power electronics to dynamically manage electricity demand, storage, and supply. They enhance the efficiency of renewable energy integration, reduce transmission losses, and improve grid stability.

Green Industries: Industries are increasingly adopting power electronics to minimize their carbon footprint. Applications include energy-efficient motor drives, power factor correction, and the integration of on-site renewable energy generation.

Electric and Hybrid Vehicles: The growth of electric and hybrid vehicles relies heavily on power electronics for battery management, motor control, and regenerative braking. These vehicles offer significant reductions in greenhouse gas emissions compared to conventional vehicles.

Electric Aircraft: Power electronics is enabling the development of electric propulsion systems for aircraft, paving the way for cleaner and quieter air travel.

Renewable Energy Systems: The percentage of renewable energy adoption is increasing significantly due to advancements in power electronics. For instance, in 2023, renewable energy sources accounted for approximately 30% of global electricity generation, with projections indicating further growth in the coming years.

Data Point 1: According to the International Energy Agency (IEA), the global capacity of solar photovoltaic (PV) systems has increased by over 25% annually in recent years, largely due to the efficiency and cost-effectiveness enabled by power electronics [5].

Data Point 2: The use of power electronics in wind turbines has improved their efficiency by 15-20%, allowing for greater energy capture and more stable output.

Bangladesh Specific Data

Data Point 3: Data from the Power Grid Company of Bangladesh (PGCB) shows that on January 5, 2025, the peak demand for electricity was 10052 MW. This demand was met through a combination of sources, including gas (4,197 MW), liquid fuel (250 MW), coal (3,265 MW), hydro (46 MW), and imports. The data underscores the need for increased renewable energy integration to reduce reliance on fossil fuels.



Figure 1: Hourly Electricity Generation and Demand in Bangladesh (January 5, 2025) [6]

Data Point 4: PGCB's data on imported power from India on the same date shows that the Bheramara HVDC station imported around 918 MW and 1000 MW, while the Tripura import was around 56 MW and 62 MW. Adani's import reached around 756 MW and 1384 MW. Power import from neighbouring countries will rise in future as many power plants are in under construction. Power electronics is mandatory for the transmission and distribution of this imported power.

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Figure 2: PGCB's Data on Imported Power from India (January 5, 2025) [6]

Data Point 5: BPDB data indicates that the installed generation capacity in Bangladesh is 27,740 MW as of in January 2025 [6]. The majority comes from thermal power plants, highlighting the potential for power electronics to improve efficiency and reduce emissions in existing infrastructure.

Installed Capacity of BPDB Power Plants as on January 2025

Fuel Type	Capacity(Unit)	Total(%)
Gas	12333 MW	44.46 %
HFO	5885 MW	21.21 %
HSD	290 MW	1.05 %
Coal	7179 MW	25.88 %
Imported	1160 MW	4.18 %
Solar	603 MW	2.17 %
Hydro	230 MW	0.83 %
Wind	60 MW	0.22 %
none	0 MW	0.00 %
Total	27740 MW	100.00 %

Figure 3: Installed Generation Capacity in Bangladesh (in January 2025) [6]

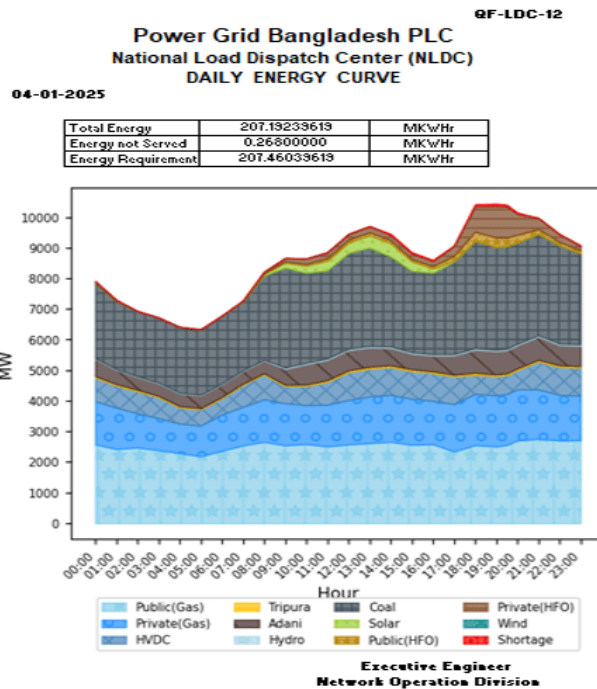


Figure 4: Daily Energy Curve of Bangladesh (January 4, 2025) [6]

Data Point 6: Figure 5 represents how Power electronics has been becoming demnading year by year in every sector because of offering accurate switching and controllability. And Figure 6 represents an alarming data that earth is becoming warmer day by day. So, usage of clean energy should be encouraged and here Power electronics can play a vital role sepecially in the power genaration section as it fulfills its duty without disturbing the Globe.

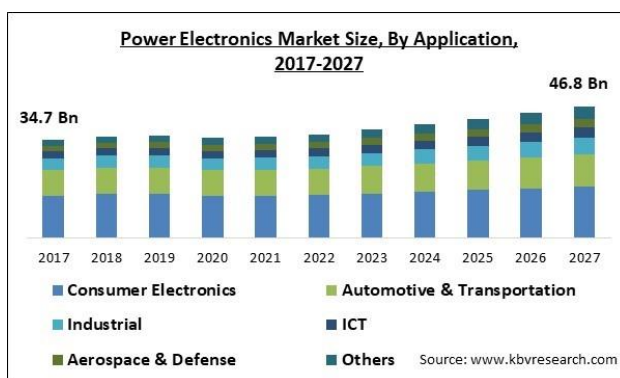


Figure 5: Power Electronics Application comparison Graph [8]

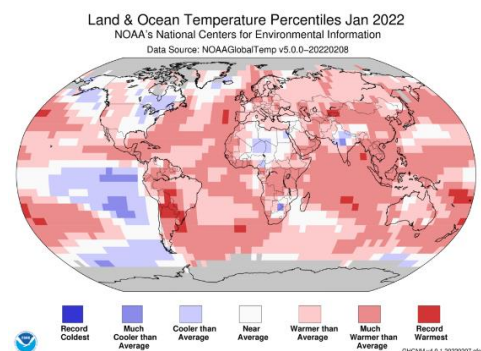


Figure 6: Land and ocean Temperature percentile [9]

Data Point 7: According to BPDB (Bangladesh Power Development Board), Bangladesh's power generation mix primarily relies on fossil fuels. However, there's a growing emphasis on renewable energy, with targets to increase its share significantly.

Future Scopes

The future of power electronics holds immense promise for advancing environmental sustainability.

Next-Generation Power Converters: Research and development efforts are focused on creating more efficient, compact, and cost-effective power converters. These converters will play a crucial role in optimizing energy use across various applications, from data centres to industrial facilities.

Wider Adoption of Electric Aircraft and Green Transportation: The transportation sector is undergoing a significant transformation with the increasing adoption of electric vehicles and the development of electric aircraft [7]. Power electronics will continue to be a key enabler of this transition, driving down emissions and reducing reliance on fossil fuels.

Advanced Grid Technologies: Smart grids will become even more sophisticated, with power electronics enabling advanced features such as self-healing capabilities, microgrid integration, and real-time optimization of energy flows.

Materials and Devices: Ongoing research into wide-bandgap semiconductors, such as silicon carbide (SiC) and gallium nitride (GaN), promises to further enhance the efficiency and performance of power electronic devices.

Bangladesh Perspective

Bangladesh has recognized the importance of transitioning to cleaner energy sources and has set ambitious targets for renewable energy adoption. The government aims to generate 40% of the country's electricity from renewable sources by 2041.

Power electronics will play a crucial role in achieving these goals:

Renewable Energy Integration: Power electronics will be essential for integrating solar, wind, and other renewable energy sources into the national grid. This will require investments in smart grid infrastructure, including advanced power converters, energy storage systems, and grid management technologies.

Energy Efficiency Improvements: Power electronics can significantly improve energy efficiency in various sectors, including industry, buildings, and transportation. This can help reduce electricity demand, lower energy costs, and minimize the environmental impact of energy consumption.

Electric Mobility: As Bangladesh considers the adoption of electric vehicles, power electronics will be crucial for charging infrastructure, battery management, and vehicle propulsion systems.

Conclusion

Power electronics is a cornerstone technology for achieving environmental sustainability. Its ability to improve energy efficiency, integrate renewable energy sources, and enable sustainable transportation systems makes it an indispensable tool in the global effort to combat climate change. The data presented in this report underscores the significant impact of power electronics across various sectors. From smart grids to electric vehicles, power electronics is driving down emissions, reducing reliance on fossil fuels, and paving the way for a cleaner, more sustainable future.

In the context of Bangladesh, power electronics offers a pathway to a more sustainable and resilient power sector. By embracing these technologies and investing in capacity building, Bangladesh can accelerate its transition to cleaner energy, improve energy access, and contribute to global efforts to mitigate climate change. The journey towards environmental sustainability requires collective action, technological innovation, and policy support. Power electronics stands as a powerful enabler in this journey, offering a brighter, cleaner, and more sustainable future for all.

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